

D. Jeff and Removing Periods

time limit per test: 3 seconds
memory limit per test: 256 megabytes

Cosider a sequence, consisting of n integers: $a_1, a_2, \dots, a_n$. Jeff can perform the following operation on sequence a :

- take three integers v, t, k ($1 \leq v, t \leq n$; $0 \leq k$; $v + tk \leq n$), such that $a_v = a_{v+t}, a_{v+t} = a_{v+2t}, \dots, a_{v+t(k-1)} = a_{v+tk}$;
- remove elements $a_v, a_{v+t}, \dots, a_{v+tk}$ from the sequence a , the remaining elements should be reindexed $a_1, a_2, \dots, a_{n-k-1}$.
- permute in some order the remaining elements of sequence a .

A beauty of a sequence a is the minimum number of operations that is needed to delete all elements from sequence a .

Jeff's written down a sequence of m integers $b_1, b_2, \dots, b_m$. Now he wants to ask q questions. Each question can be described with two integers l_i, r_i . The answer to the question is the beauty of sequence $b_{l_i}, b_{l_i+1}, \dots, b_{r_i}$. You are given the sequence b and all questions. Help Jeff, answer all his questions.

Input

The first line contains integer m ($1 \leq m \leq 10^5$). The next line contains m integers $b_1, b_2, \dots, b_m$ ($1 \leq b_i \leq 10^5$).

The third line contains integer q ($1 \leq q \leq 10^5$) — the number of questions. The next q lines contain pairs of integers, i -th of them contains a pair of integers l_i, r_i ($1 \leq l_i \leq r_i \leq m$) — the description of i -th question.

Output

In q lines print the answers to Jeff's queries. Print the answers according to the order of questions in input.

Examples

input

Copy

5
2 2 1 1 2
5
1 5
1 1
2 2
1 3
2 3

output

Copy

2
1
1
2
2

input

Copy

10
2 1 3 3 3 3 1 3 1 1
10
4 8
2 10
1 10
4 4
1 3
2 4
6 7
1 9
2 5
1 1

output

Copy

2
3
3
1
3
2
2
3
2
1

Codeforces Round 204 (Div. 1)

Finished

Practice

Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

Last submissions

Submission	Time	Verdict
364406993	Feb/26/2026 00:55	Accepted

Problem tags

data structures

*2700

No tag edit access

Contest materials

Codeforces Round #204

Tutorial